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Advisory Automated System for Wellbore Cleaning and Evaluation Improves Drilling Efficiency

M. M. Al-Rubaii, ARAMCO, DHAHRAN, EASTREN PROVINCE, KSA

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Abstract

Ineffective hole cleaning leads to a complicated drilling hole problems such as stuck pipe incidents, and difficult tripping operations. The objective of this paper to introduce a unique Hole Cleaning Advisory System which is showing six real time hole cleaning indicators that can provide an obvious vision on hole cleaning performance during drilling operations and thereby improves drilling efficiency. They are real time drilling fluid density (MWeff), real time equivalent circulating density (ECDeff), real time developed carrying capacity index (HCI), and real time developed cuttings concentration in annulus (CCA), real time developed transport ratio (CTRM) and real time transport index (TIm). Artificial intelligence tool such as artificial neural network (ANN) was applied for validation and confirmation of models parameters and obtaining similar real time profiles of hole cleaning indices. The system was utilizing rig sensors' real time values, surface and field operations readings by collecting historical and real time readings , selecting the most relevant and important factors.

Performing data analytics, preparation and interpretation. Finding the relationships between parameters. Modifying critical parameters. Validating models with referenced tools those are applicable and utilized in drilling operations. Integrating developed models in real time drilling operations to evaluate and monitor drilling performance. The conducted process, validations of real time models were done and compared with other commercial software and advanced drilling tool such as Pressure while drilling (PWD) tool and other sensors tools provided by Services Company for drilling operations. The system were applied in the fields and provided a firm and solid validations of the developed system that empowered the new technology to be applied in drilling rigs fleet.

The results showed that the MWeff and ECDeff were validated with pressure while drilling (PWD) and resulted in 2 % errors. MWeff was validated with other commercial software shows real time mud as output, the developed real time model of mud weight while drilling showed errors 2 %. That means the developed models for mud weight and equivalent circulating density were robust and can be utilize while drilling operation to optimize well drilling performance. regarding other hole cleaning indices such as HCI, CCA, CTRM & TIm were applied in several types of well, it was able to identify the hole cleaning issues and improves the drilling efficiency specifically rate of penetration (ROP) by 65 %. The drilling engineering and operation team was performing immediate intervention to avoid drilling troubles due to hole cleaning.

The final results contributed for optimizing drilling efficiency significantly in safety and environmental manners with cost and time effectiveness. In addition, participating in drilling automation and digitalization in compatibility with fourth industrial revolution (4-IR).

Introduction

Inadequate hole cleaning critically impedes drilling efficiency, causing costly NPT events like stuck pipe (Ogunrinde and Dosunmu, 2012; A. A. Mahmoud et al., 2020). Effective cuttings transport, influenced by flow rate, mud rheology, and pipe rotation, is paramount, especially given the impact of inclination angle on cuttings bed formation (Gaurina-Medjimurec and Pasic, 2014; Larry W, 2007). Existing borehole cleaning models (e.g., HCR, CCI, CCA, TR, TI) suffer from real-time applicability limitations and incomplete parameter integration (Rasi, 1994; Robinson and Morgan, 2004; Newitt et al., 1961), highlighting a persistent industry challenge for robust, real-time, integrated solutions.

Machine Learning (ML) empowers data-driven adaptation, discerning variable dependencies without explicit programming (Gul, 2021; Badillo et al., 2020). This core Artificial Intelligence (AI) component is crucial for real-time optimization, guiding variable selection and model deployment (Bangert, 2021). Real-time drilling fluid property measurement is critical for optimizing operations and mitigating costly NPT (Liu et al., 2021b). Managing Equivalent Circulating Density (ECD) is paramount in extended-reach wells, as fluctuations cause instability (Abdelaal et al., 2022; Roy et al., 2021).

Consequently, AI techniques like Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Decision Trees (DT) are increasingly used for real-time prediction of drilling fluid properties and wellbore conditions (Alkinani et al., 2019b, 2020; Hashemizadeh et al., 2021; J. Wang et al., 2022; Ahmadi, 2016; Alsaihati et al., 2021; Gamal et al., 2021; Rahmati and Tatar, 2019).

Their efficiency can be optimized by refining input parameters such as SPP, ROP, RPM, mud properties, and LSYP. This work introduces a unique Hole Cleaning Advisory System with six real-time indicators to optimize Rate of Penetration (ROP) and mitigate NPT, addressing the lack of a comprehensive, validated real-time system. Objectives include developing novel models for Transport Index (TIm), Hole-Cleaning Index (HCI), Cutting Transport Ratio (CTRM), Equivalent Circulating Density (ECDeffc.m), and Effective Mud Weight (MWeffc.m). These models will be rigorously validated through offline analysis, comparison with real-time surface sensors (commercial technology), validation against real-time downhole sensors (PWD), and comparative analysis against AI techniques for predicting drilling fluid density, ECD, TIm, HCI, CCA, and CTRM.

Literature Review

Optimizing downhole cleaning is critical for drilling efficiency, reducing NPT, and improving cementation (Ogunrinde and Dosunmu, 2012). Inadequate cleaning causes high ECD, torque/drag, instability, and stuck pipe (A. A. Mahmoud et al., 2020; Mitchell, 2003). Hole cleaning dynamics vary with wellbore inclination; pipe rotation is vital in deviated sections to prevent cuttings bed formation (Gaurina-Medjimurec and Pasic, 2014; Larry W, 2007). Cuttings settling velocity, density, and size are key removal parameters (Agwu et al., 2018).

Methods to enhance cleaning include optimizing drilling fluid properties, BHA design, hydraulics, and rig systems (Khosravanian and Aadnøy, 2021a). Fluid rheology (PV, YP, gel strength, density) is crucial for cuttings transport and wellbore stability (Al-Shargabi et al., 2022; S. Davoodi et al., 2021). Pipe rotation promotes turbulent flow (Adari et al., 2000; Williams and Bruce, 1951), and ROP management controls cuttings load (Unegbu, 2010).

Water-Based Muds (WBMs) are preferred for their environmental and cost benefits, performing critical functions beyond cuttings removal (Bridges and Robinson, 2020). Non-Newtonian fluids offer superior carrying capacity (Adjei et al., 2023), and solid control equipment maintains optimal mud rheology (Al-

Rubaii et al., 2023c). Rheological properties profoundly impact hole cleaning: high PV/AV enhance suspension, YP aids static suspension, and gel strength is crucial during circulation stops (Kelin et al., 2013). Chemical additives (viscosifiers, inhibitors, rheology modifiers) optimize mud properties, minimize wellbore instability, and improve efficiency (Humood et al., 2019; Khosravanian and Aadnøy, 2021b). Research consistently shows positive impacts of various additives (nanoparticles, polymers) on rheology and fluid loss (Aberoumand et al., 2016; Alsaba et al., 2020; Amorin and Bediako, 2020; Özkan, 2018; Sargani et al., 2020).

Despite existing models (HCR, CCI, CCA, TR, TI), many lack real-time applicability, dynamic mud weight consideration, or accurate cuttings properties (Jimmy et al., 2022; Khosravanian and Aadnøy, 2021a). This necessitates an advanced, real-time, integrated advisory system for proactive wellbore cleaning.

Methodology and Development

This section details the advanced methodology of our novel Hole Cleaning Advisory System, integrating sophisticated modeling with AI for real-time wellbore evaluation. Its core comprises five mathematically robust, newly developed indicators: TIm, HCI, CTRm, ECDeffc.m, and MWeffc.m. These models uniquely advance beyond conventional approaches by integrating dynamic drilling conditions, comprehensive parameter sets (e.g., dynamic mud weight, Angle Factor, Rheology Factor, LSYP, cuttings concentration, hole geometry, CRF), and AI-driven refinements. Artificial Intelligence (ANN, DT, and SVM) was strategically employed for rigorous validation and real-time prediction, using extensive offshore well data (4,371 ECD, 33,588 MW records). ANN models consistently excelled for TIm, CTRm, CCAm, and HCI predictions (e.g., CCAm at 0.9996 correlation), ensuring the system's accuracy, robustness, and real-time operational applicability.

Field Applications Using The Novel Models

This section details the extensive field validation of our novel TIm, HCI, CTRm, ECDeffc.m, and MWeffc.m models, demonstrating their transformative impact on drilling. Rigorously tested in offshore wells (Well-A, Well-B, Well-C) across diverse conditions (30-90° inclination, varied formations), these models offer unprecedented real-time prediction and control.

TIm (Transport Index Modified): Showed enhanced performance with increasing MWeff and LSYP. In Well-A, TIm indicated excellent cleaning (Figure 1a), leading to a 56% ROP improvement and 44% torque reduction vs. Well-B (Figure 2a,b,c,d; Table 1). It also prevented a stuck pipe in Well-C (Figure 3a,b,c; Table 2), providing instant, actionable predictions. HCI (Hole-Cleaning Index): Validated in Well-A, B, and C (Figures 4, 5, 6), HCI consistently demonstrated superior accuracy over CCI, enabling proactive ROP adjustments and preventing stuck pipe (Figure 4a, 5, 5a, 6; Table 4).

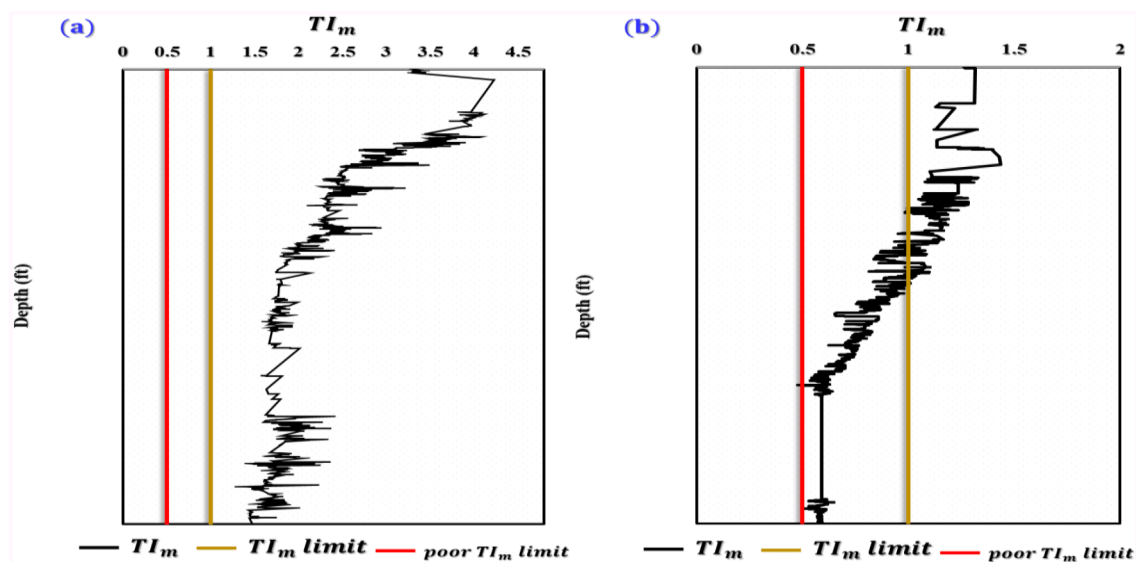


Figure 1—Application of TI_m in offshore Well-A with proper borehole cleaning (a) and Well-B with poor borehole cleaning (b).

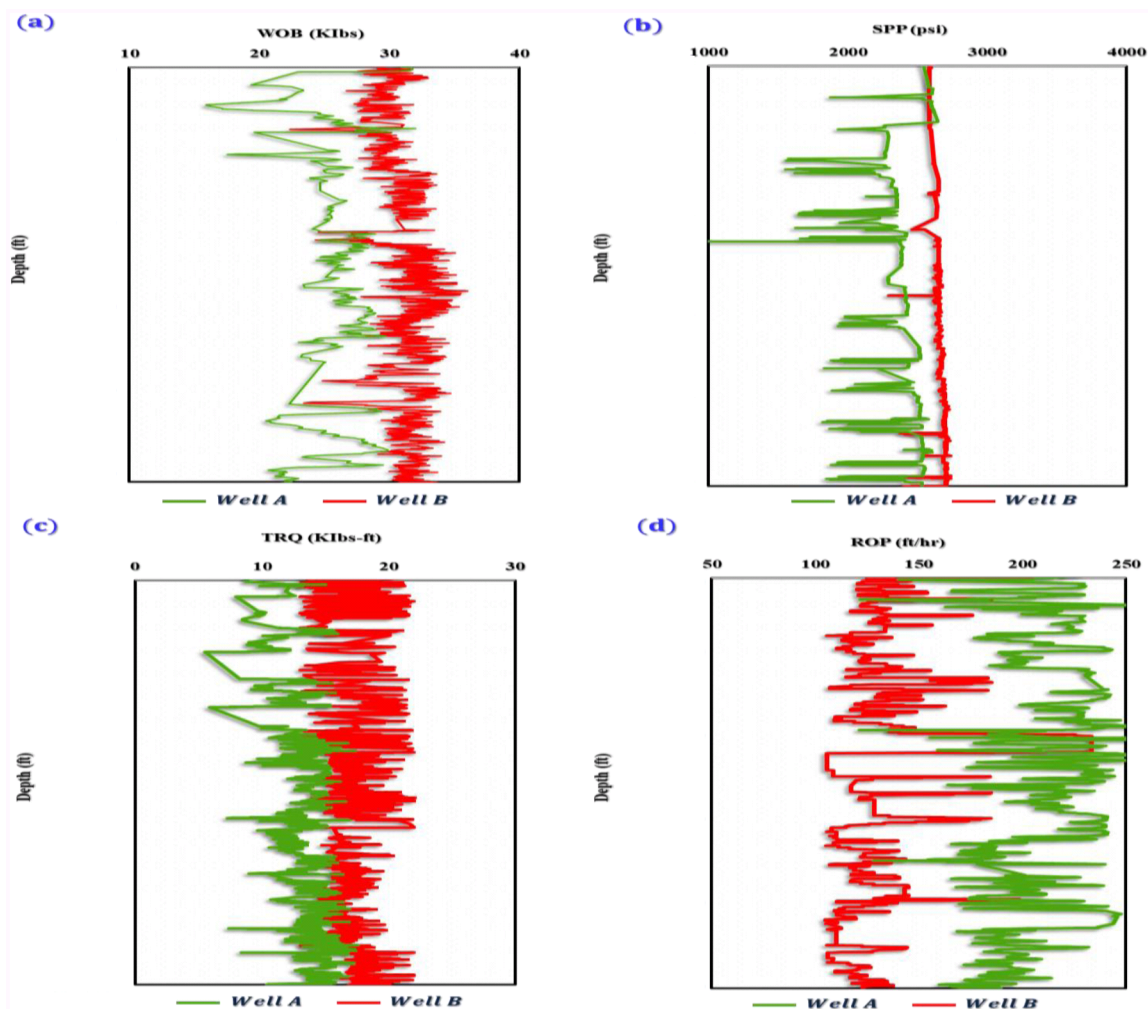


Figure 2—The changes in drilling parameters for Well A with proper borehole cleaning and Well B with poor borehole cleaning: (a) WOB, (b) SPP, (c) TRQ, and (d) ROP.

Table 1—Effect of using the novel TI_m model on well performance.

Effect of using the novel TI_m model on the performance of Well-A					
Nº	Items (output)	Minimum	Maximum	Average	Statement
1	TI_m	1.4	4	2.4	Effective borehole cleaning
2	ROP	120	280	209	Optimized ROP by 56% due to the effective borehole cleaning
3	TRQ	5	18	9.6	Decreased TRQ by 44% due to the effective borehole cleaning
Effect of using the novel TI_m model on the performance of Well-B					
Nº	Items (output)	Minimum	Maximum	Average	Statement
1	TI_m	0.47	1.4	0.79	Insufficient borehole cleaning
2	ROP	105	258	134	Lower ROP due to the insufficient borehole cleaning
3	TRQ	13	22	17	Higher TRQ due to the insufficient borehole cleaning and cutting accumulation

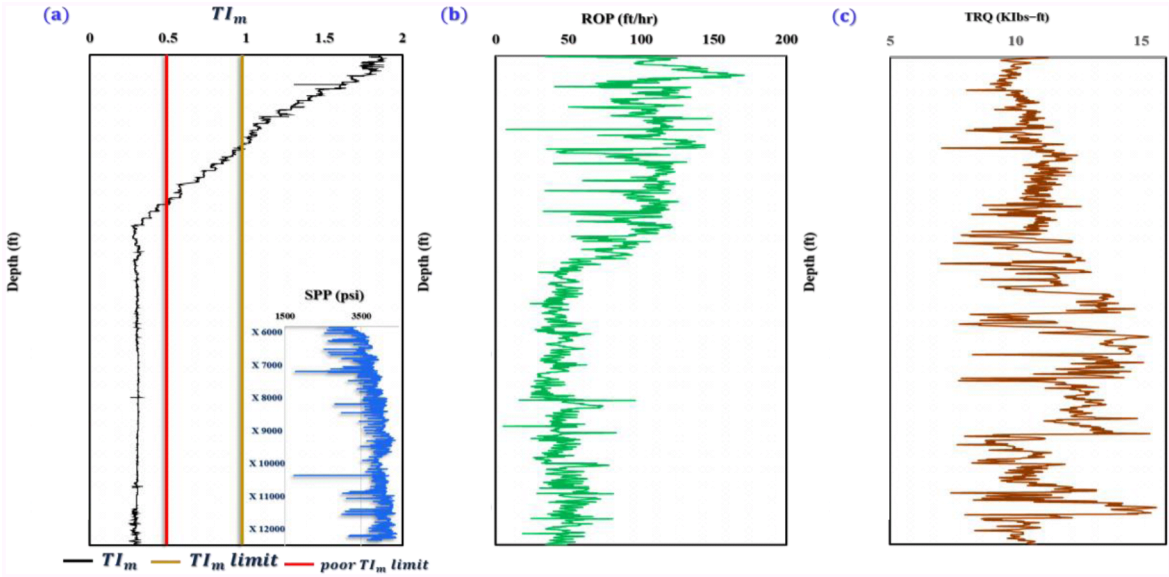


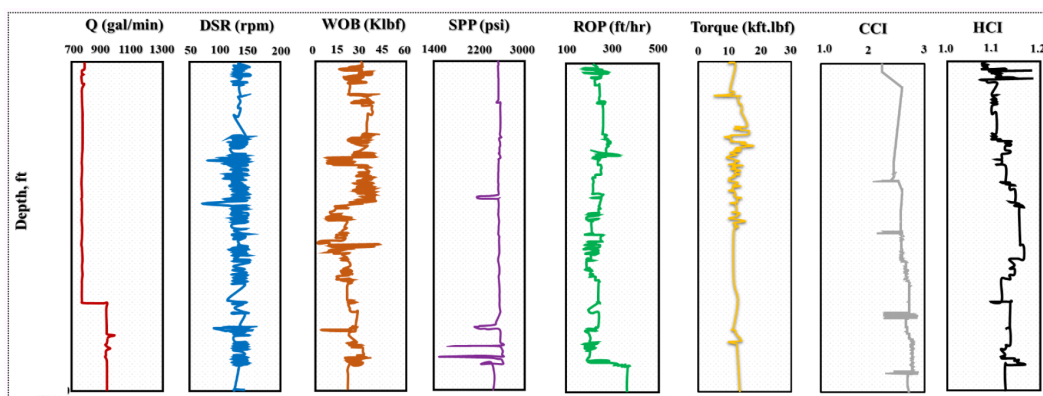
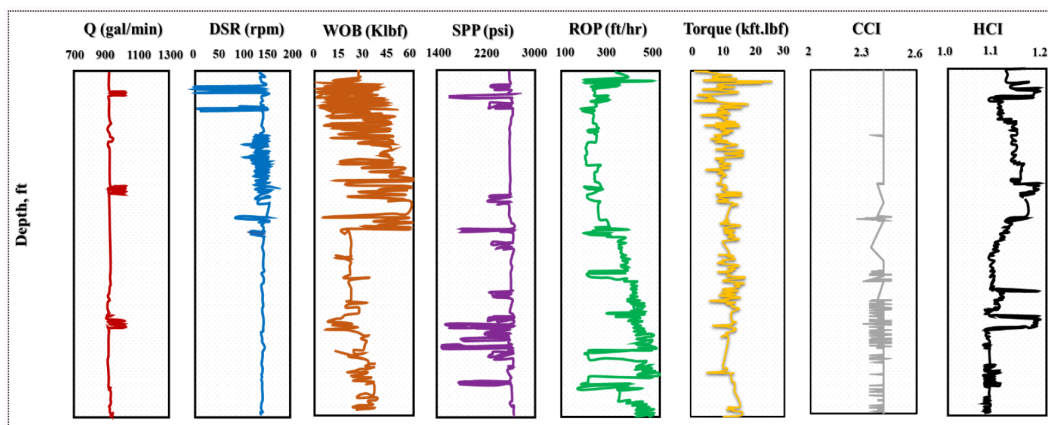
Figure 3—Application of TI_m in Well C in the case of a stuck pipe:
(a) Well C with poor borehole cleaning & SPP, (b) ROP, and (c) TRQ.

Table 2—Effect of utilizing the novel TI_m model on well performance.

Effect of using the novel TI_m model on the performance of Well-C					
Nº	Items (output)	Minimum	Maximum	Average	Statement
1	TI_m	0.25	1.8	0.56	Insufficient borehole cleaning
2	ROP	4.92	166	93.3	Lower ROP due to insufficient borehole cleaning
3	TRQ	7.24	16	12.5	Higher TRQ due to the cutting accumulation resulted in stuck pipe incident

Table 3—Effect of utilizing the novel HCI model on well performance

Performance of Well-A employing <i>HCI</i>					
Items	Output	Minimum	Maximum	Average	Remark
1	<i>HCI</i>	0.8	1.9	1.5	Optimized hole cleaning efficiency
2	<i>CA</i>	0.012	0.039	0.024	Smooth cuttings accumulation in annulus removal due to optimized hole cleaning efficiency
3	<i>ROP</i>	120	280	205	Optimized drilling performance due to proper hole cleaning efficiency
Performance of Well-B without employing <i>HCI</i>					
Items	Output	Minimum	Maximum	Average	Remark
1	<i>HCI</i>	0.3	1.3	0.81	Improper hole cleaning efficiency
2	<i>CA</i>	0.03	0.08	0.04	low removal of cuttings accumulation in annulus due to improper hole cleaning efficiency
3	<i>ROP</i>	100	260	135	low drilling performance due to insufficient hole cleaning efficiency

Figure 4—Changes in *HCI* and the drilling parameters for Well-A.Figure 4a—Changes in *HCI* and the drilling parameters for Well-A.

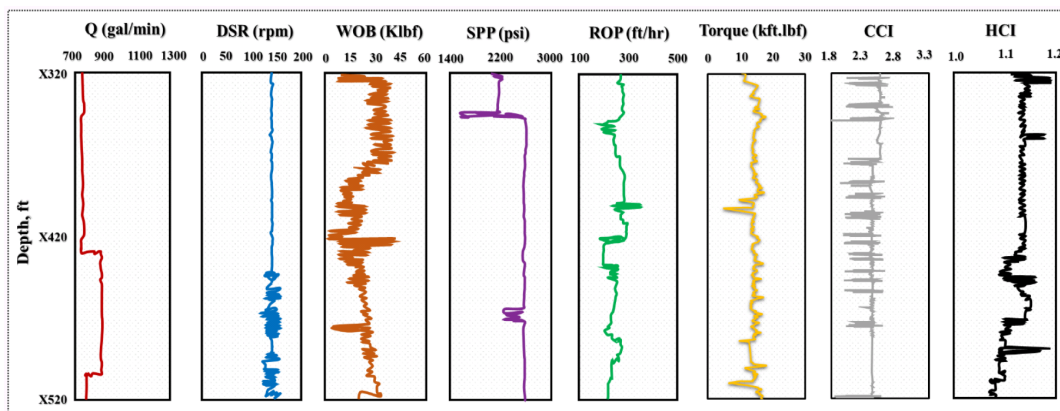
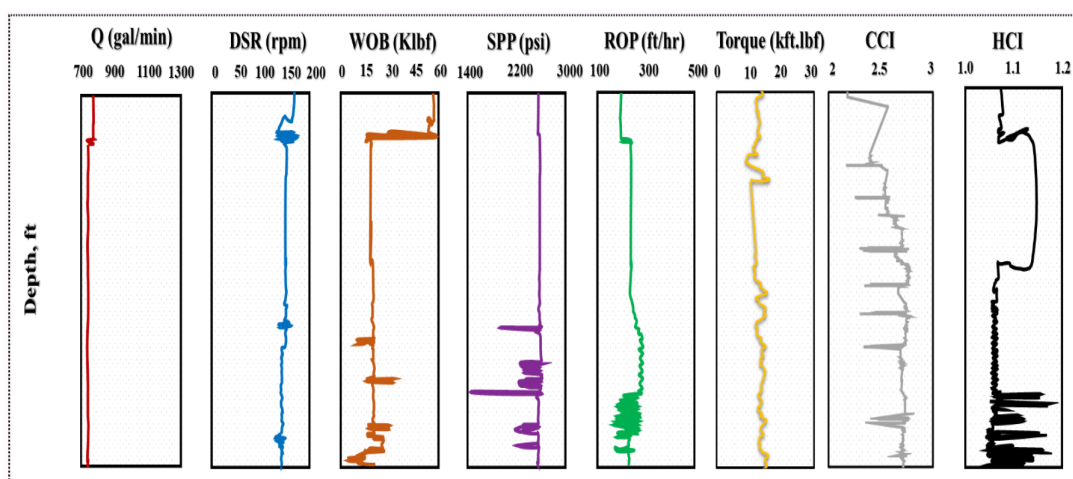
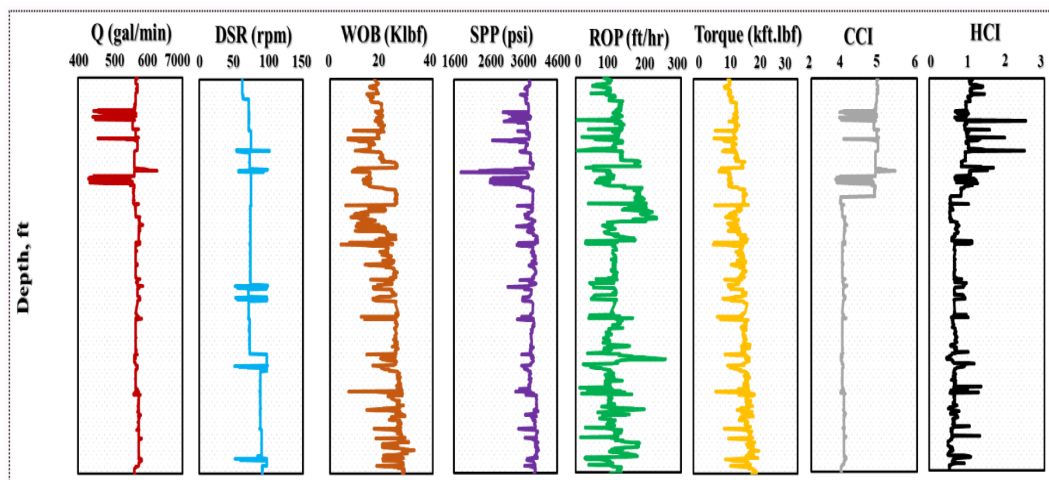
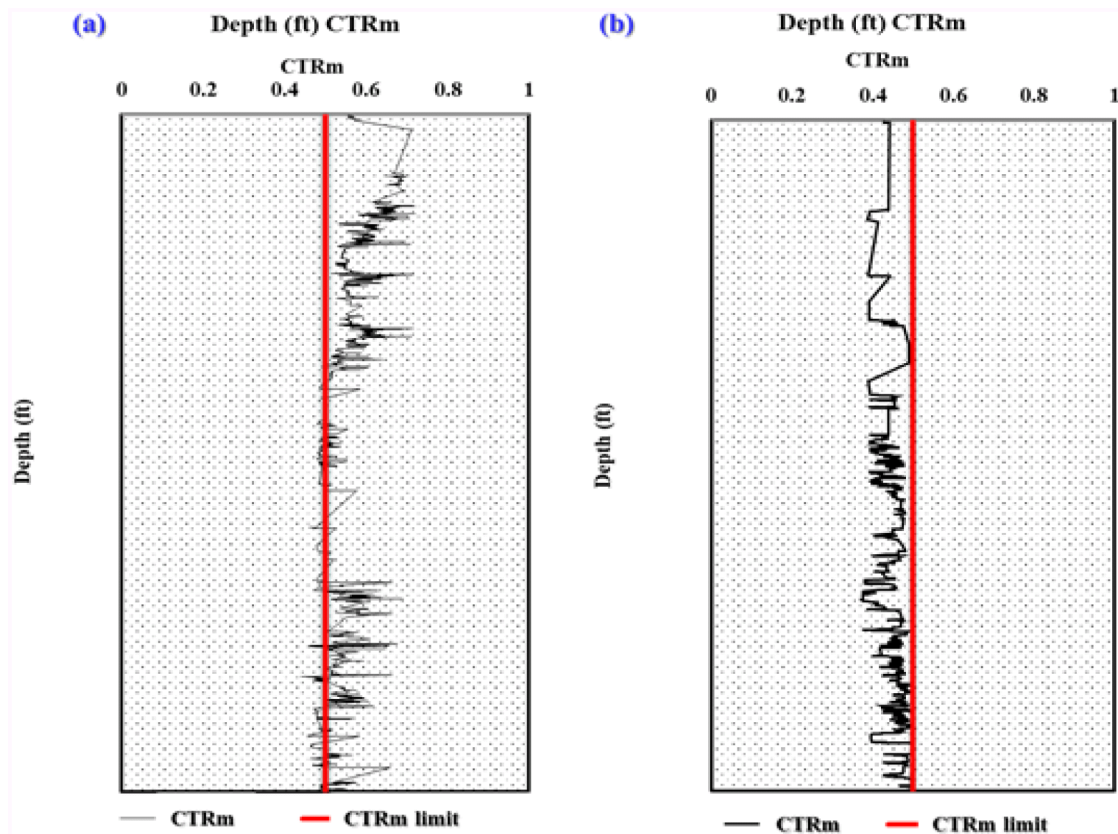
Figure 5—Changes in *HCI* and the drilling parameters for Well-B.Figure 5a—Changes in *HCI* and the drilling parameters for Well-B.Figure 6—Changes in *HCI* and the drilling parameters for Well-C.

Table 4—Effect of utilizing the novel HCI model on well performance

Performance of Well-C employing HCI					
Items	Output	Minimum	Maximum	Average	Remark
1	HCI	0.34	1.7	0.79	Improper hole cleaning efficiency
2	CA	0.02	0.2	0.064	low removal of cuttings accumulation in annulus due to improper hole cleaning efficiency
3	ROP	0.665	256	110	low drilling performance due to insufficient hole cleaning efficiency and stuck pipe

Its comprehensive nature integrates key rheological and mechanical factors. CTR_m (Cuttings Transport Ratio Modified): Validated across four wells (deviated, vertical, horizontal), CTR_m proved effective in optimizing borehole cleaning (Figures 7a, 7b, 8a, 8b; Tables 5, 6), enhancing drilling efficiency. Overall, these models provide real-time, accurate insights for proactive intervention, optimizing drilling parameters, significantly enhancing ROP, mitigating major problems, and contributing to substantial cost and time savings.

Figure 7—Application of CTR_m in the offshore proper hole cleaning of Well-A (a) and the poor hole cleaning of Well-B (b).

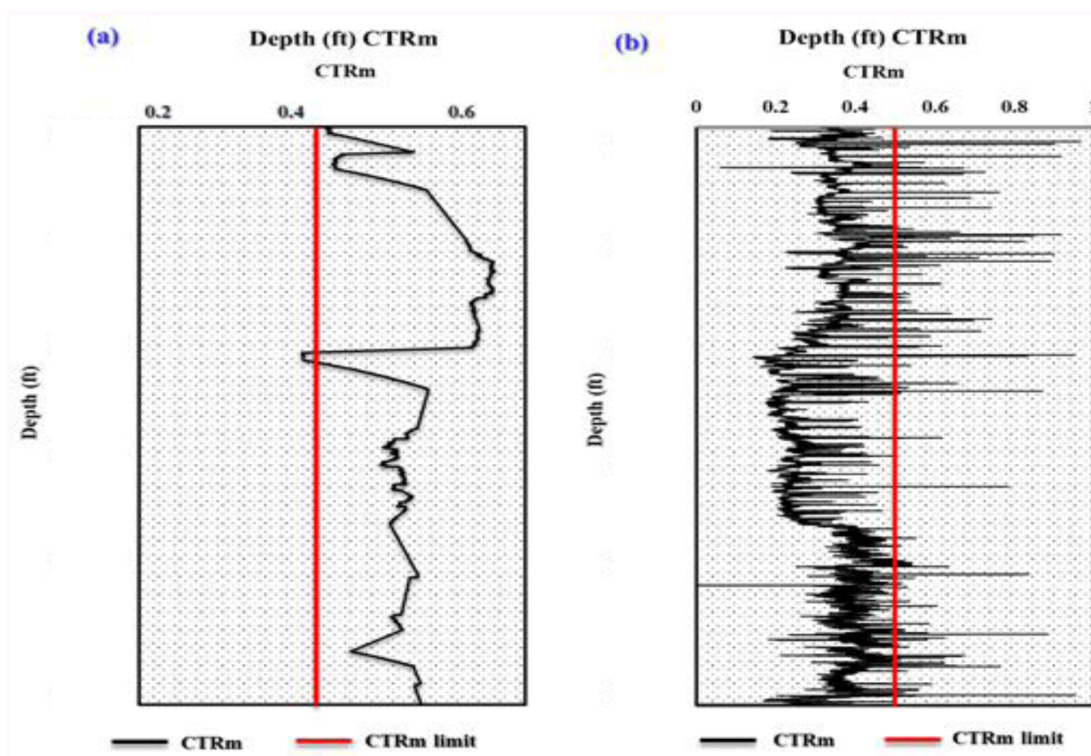


Figure 8—Application of CTR_m in the offshore gas vertical well-A (a) and the horizontal Well-B (b).

Table 5—Impact of employing CTR_m on well performance.

Performance of Well-A Employing CTR _m					
Items	Output	Minimum	Maximum	Average	Remark
1	CTR _m	0.49	0.79	0.64	Effective hole cleaning
2	ROP	120	280	205	Improved the drilling performance due to optimized efficiency of hole cleaning
Performance of Well-B employing CTR _m					
Items	Output	Minimum	Maximum	Average	Remark
1	CTR _m	0.39	0.49	0.44	Improper removal of cuttings from the wellbore
2	ROP	105	264	135	Decreased drilling efficiency and performance attributable to inefficient wellbore cleaning

Table 6—Impact of employing CTR_m on well performance.

Performance of gas Well-A Employing CTR_m					
Items	Output	Minimum	Maximum	Average	Remark
1	CTR_m	0.48	0.63	0.57	Proper removal of cuttings from the wellbore
2	ROP	5.0	25.7	14.3	Normal drilling efficiency and performance due to proper wellbore cleaning
Performance of horizontal Well-B employing CTR_m					
Items	Output	Minimum	Maximum	Average	Remark
1	CTR_m	0.58	0.95	0.45	Poor removal of cuttings from the wellbore
2	ROP	0.552	267	93.4	Decreased drilling efficiency and performance attributable to inefficient wellbore cleaning and stuck pipes

Artificial Intelligence Applications Using Novel Models

This section details the strategic integration of AI with our novel ECDeffc.m, MWeffc.m, TIm, CTR_m , and HCI models to maximize drilling efficiency via real-time prediction and automated control. Our approach overcomes existing literature's limitations (Alsaihati et al., 2021; Zheng et al., 2017; Hashemizadeh et al., 2021) by developing novel ECDeffc.m and MWeffc.m models that comprehensively integrate critical parameters (SPP, ROP, RPM, mud properties, LSYP, hole geometry). Rigorous application of ANN, SVM, and DT on extensive field data (4,371 ECD, 33,588 MW records) validates our models' remarkable accuracy.

Field applications of ECDeffc.m and MWeffc.m closely match AI predictions and real-time PWD data (Figure 9a, 9b), often surpassing AI due to direct output capability. MWeffc.m aligns with commercial MW readings (Figure 10), and ECDeffc.m is indistinguishable from established systems (Figure 11, 12), confirming their robustness. This real-time prediction is paramount for preventing formation damage, instability, and well control incidents, optimizing hole cleaning.

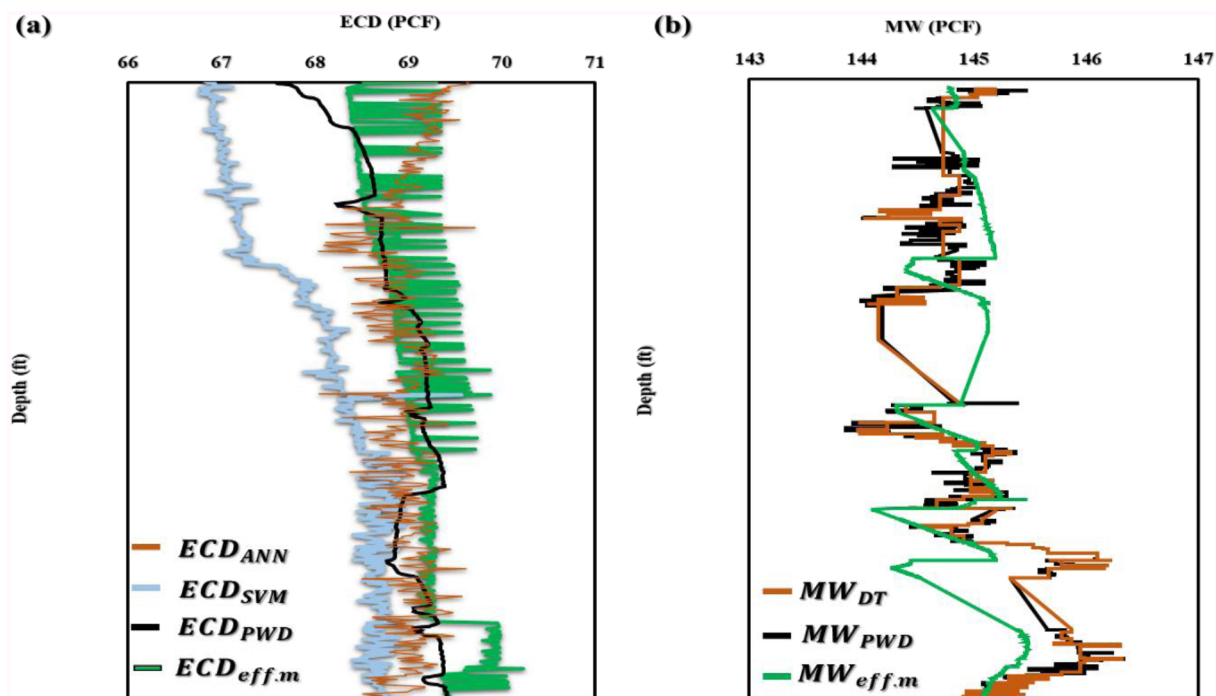


Figure 9—Comparison of field applications with AI applications using the novel model $ECD_{eff.m}$ (a) and $MW_{eff.m}$ (b)

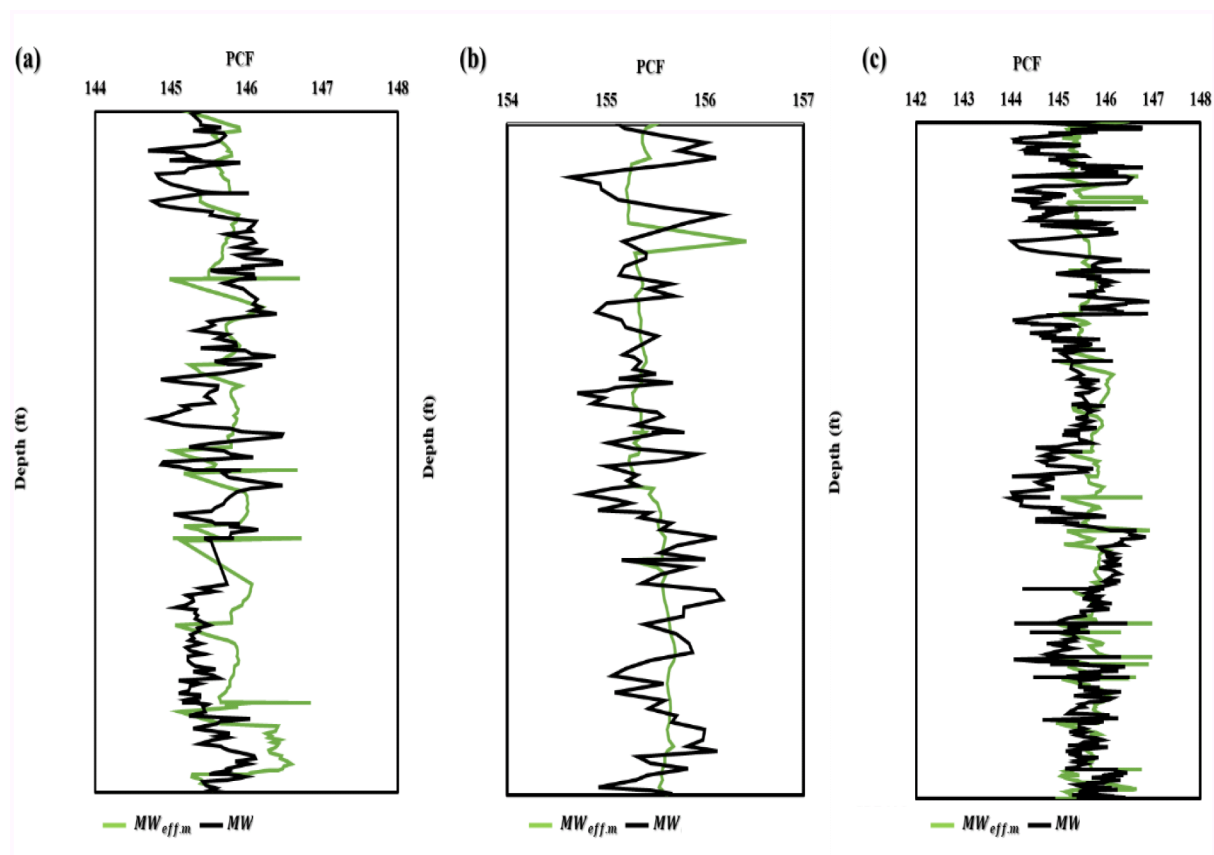


Figure 10—The comparison of the $MW_{eff.m}$ and Commercial technology MW in real time: (a) Well-A, (b) Well-B, (C) Well-C.

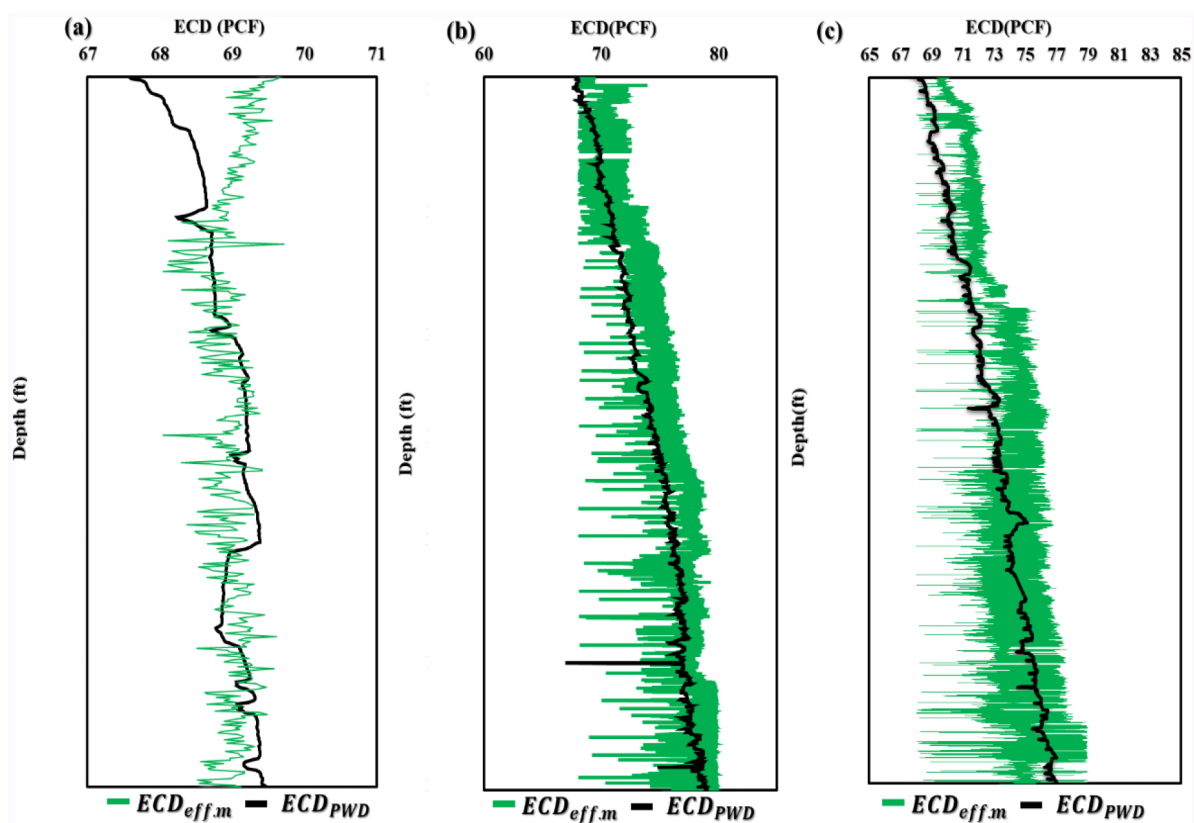


Figure 11—Application of $ECD_{eff.m}$ in offshore gas Deviated Well-A (a), offshore oil Deviated Well-B (b) and horizontal oil Well-C (c)

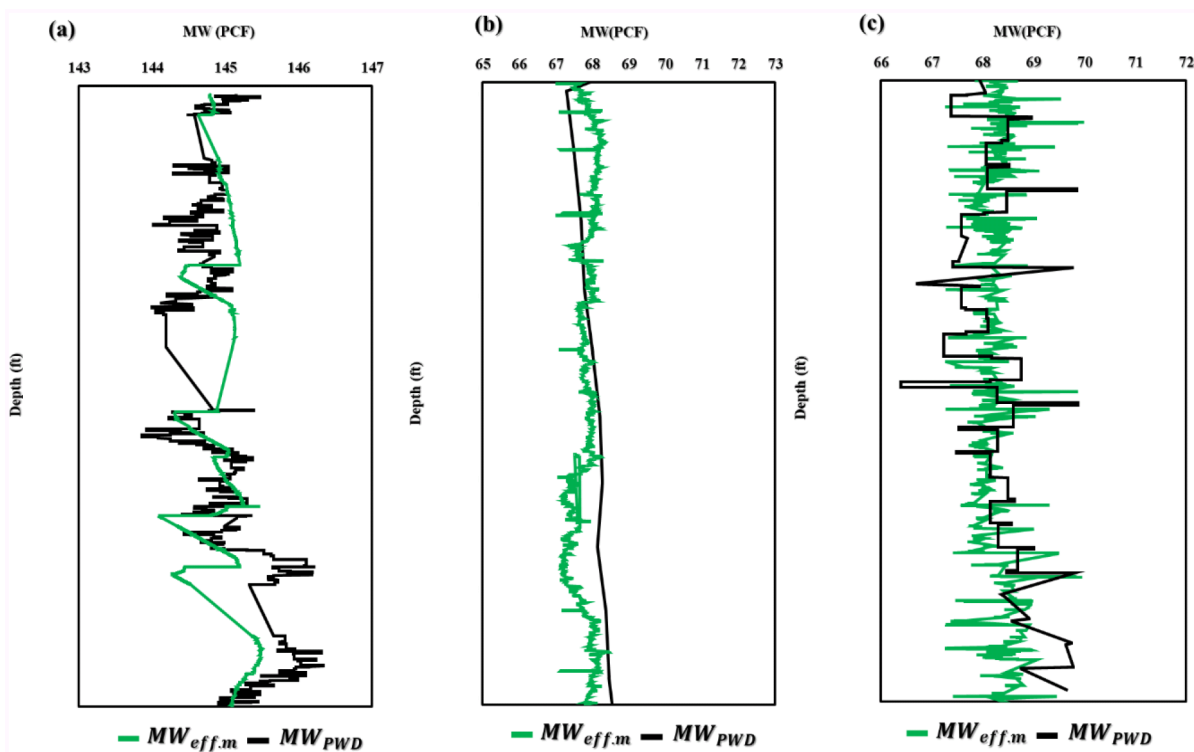


Figure 12—Application of $MW_{eff.m}$ in offshore gas Deviated Well-A (a), offshore oil Deviated Well-B (b) and horizontal oil Well-C (c).

Integrating AI with these novel, field-validated models creates a superior, proactive advisory system, enabling precise forecasts and timely adjustments that significantly enhance safety, reduce downtime, and drive cost savings in complex drilling environments. The automated system supports consistent data capture, remote oversight, and continuous improvement, accelerating industry digitalization.

AI Applications and CFD for Hole Cleaning Optimization

This section showcases the successful integration of AI and CFD for advanced hole cleaning. Our novel AI models (CCAm, TIm, HCI, CTRm) demonstrate exceptional predictive accuracy (R^2 values from 0.9138 to 0.9996) for real-time indicators (Figures 13-16). Complementing AI, Computational Fluid Dynamics (CFD) simulations offer a comprehensive understanding of drilling parameter effects on cuttings transport (Figures 17-20). CFD reveals: Fluid Velocity: Increasing velocity significantly reduces cuttings volume fraction (e.g., 60% reduction with 1-2 m/s increase) (Figure 16). Diameter Ratio: Higher ratios ($\kappa=0.4$ to 0.8) reduce cuttings volume fraction by 40% at low velocities (Figure 17). Flow Behaviour Index (n): Newtonian-like fluids (higher n) improve cuttings transport in horizontal annuli (Figure 18). Pipe Rotation & Eccentricity: Inner pipe rotation substantially reduces cuttings volume fraction (31-44% for $E=0.25$ -0.75) due to enhanced erosion (Figure 19). This integrated approach combines highly accurate AI prediction with detailed CFD insights, creating a powerful framework for optimizing wellbore cleaning, enabling proactive decision-making, and enhancing overall drilling efficiency.

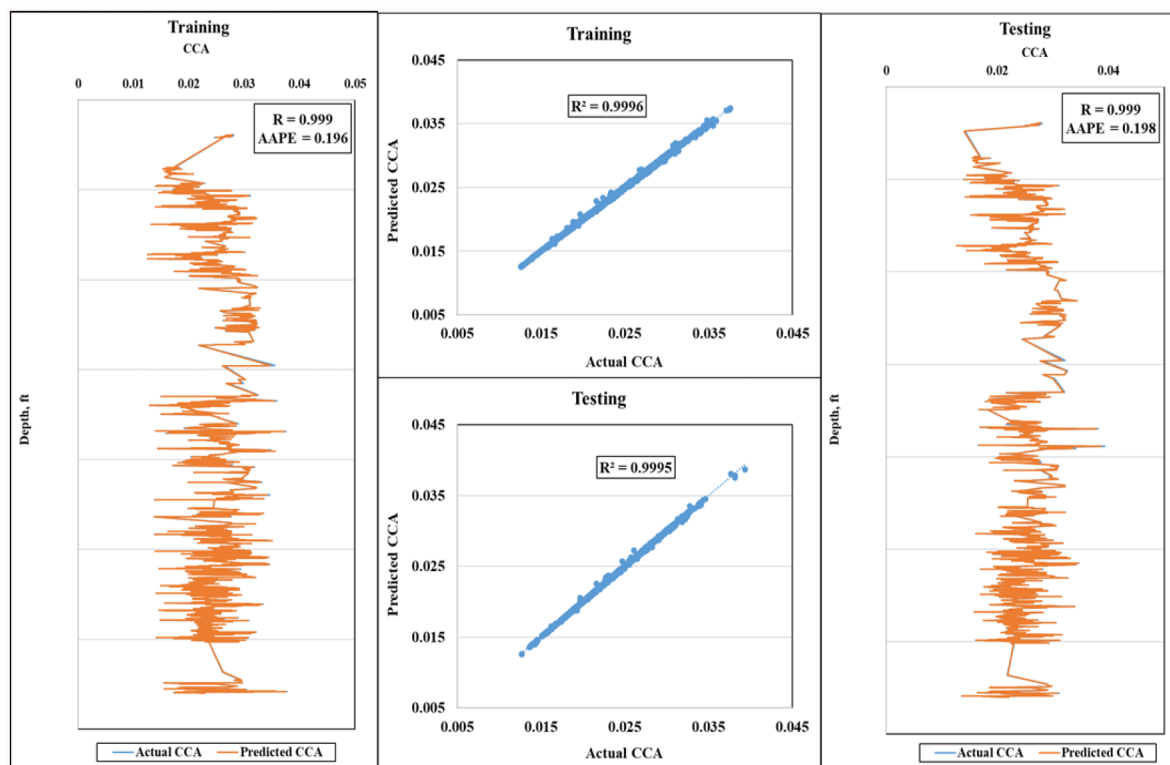


Figure 13—Comparison of field applications with AI applications using the novel model CCA: Training and Testing prediction with coefficient of determination (R^2)

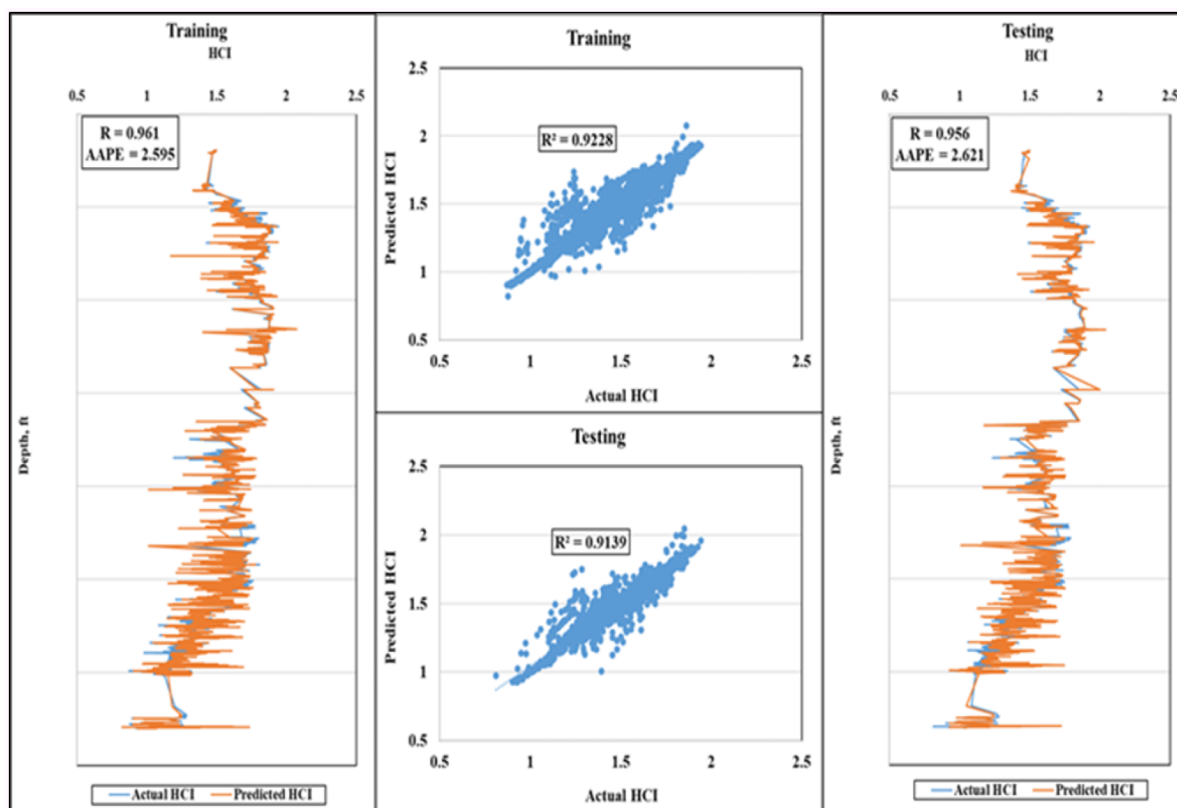


Figure 14—Comparison of field applications with AI applications using the novel model HCI: a); Training and Testing prediction with coefficient of determination (R^2)

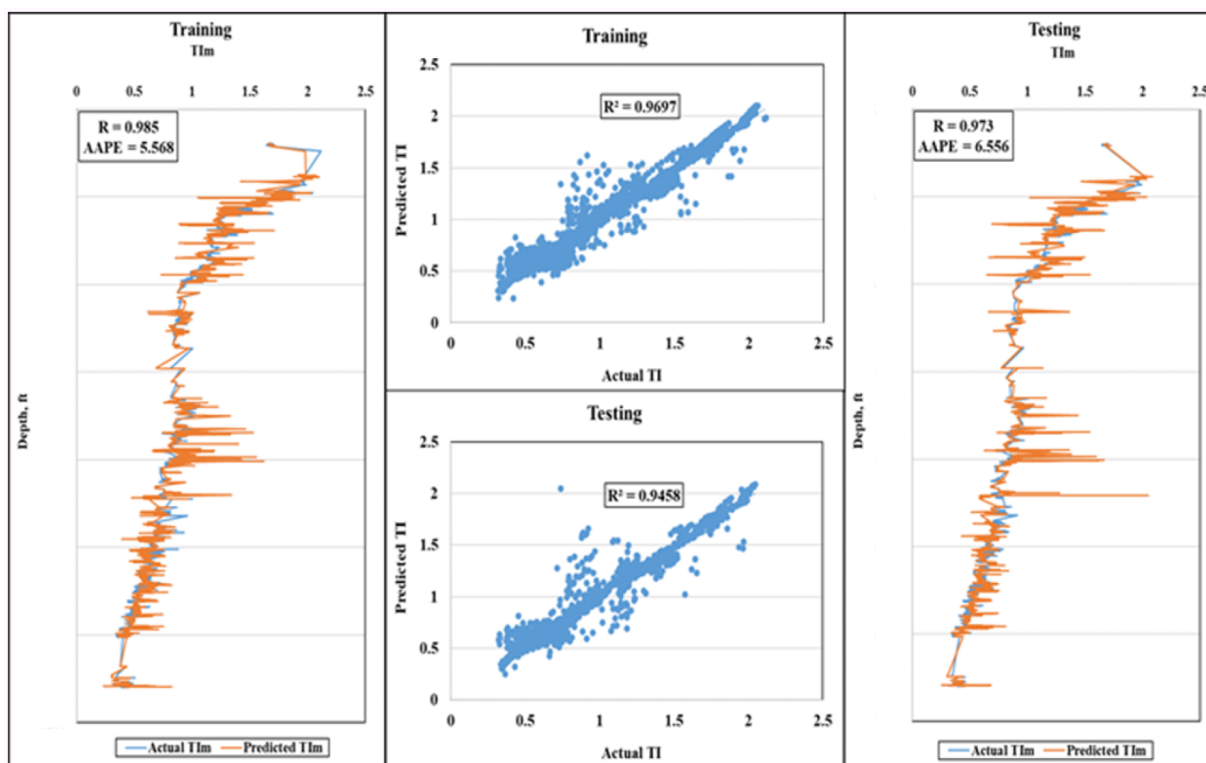


Figure 15—Comparison of field applications with AI applications using the novel model TI_m : a); Training and Testing prediction with coefficient of determination (R^2)

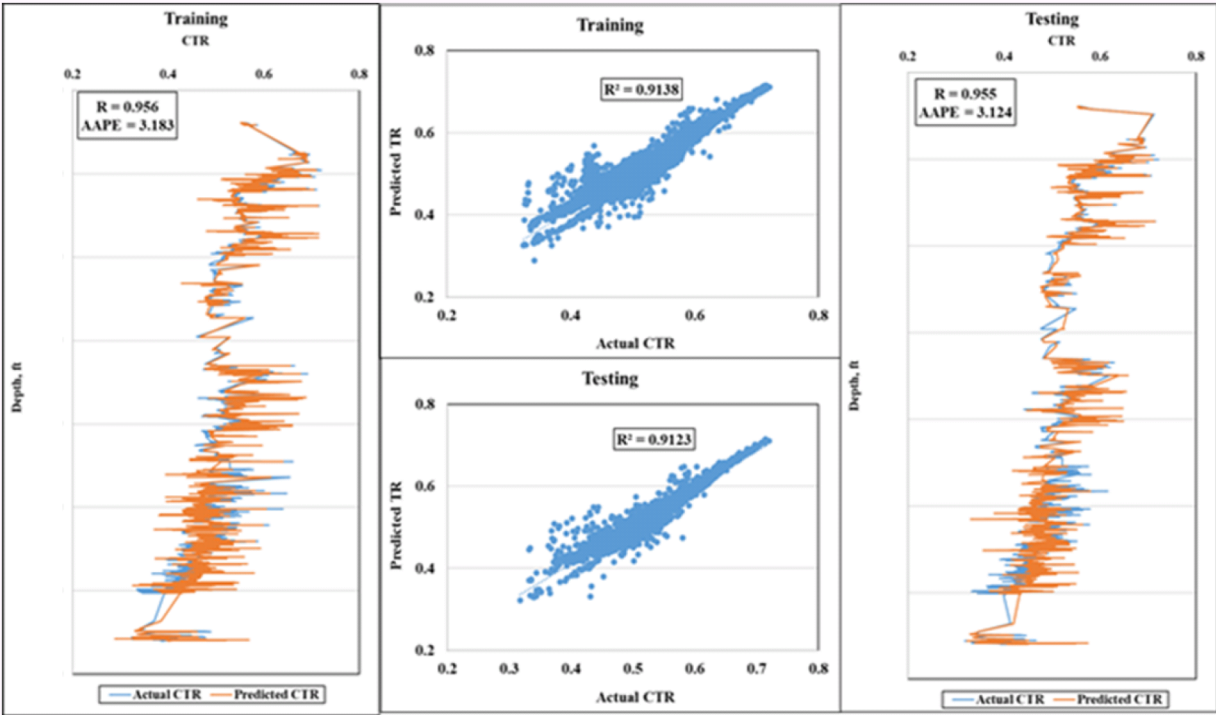


Figure 16—Comparison of field applications with AI applications using the novel model CTR_m : a); Training and Testing prediction with coefficient of determination (R^2)

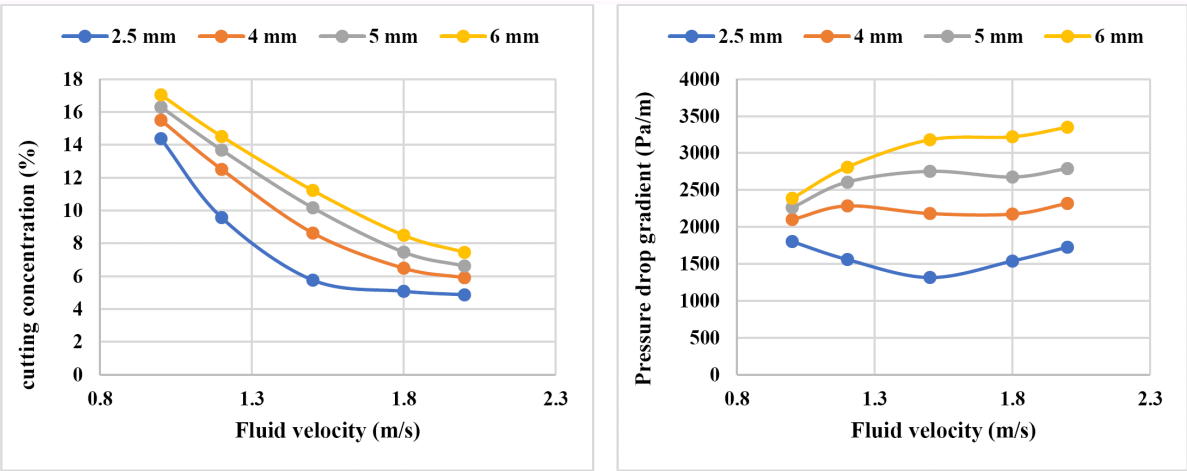


Figure 17—Influence of fluid velocity on cutting concentration and pressure gradient for various particle diameter

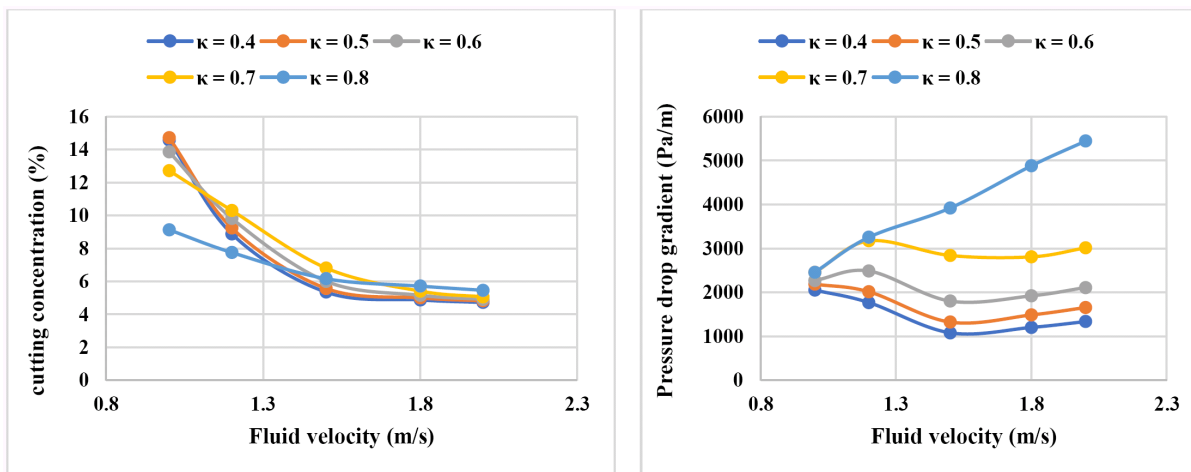


Figure 18—Influence of fluid velocity on cutting concentration and pressure gradient for various diameters ratios

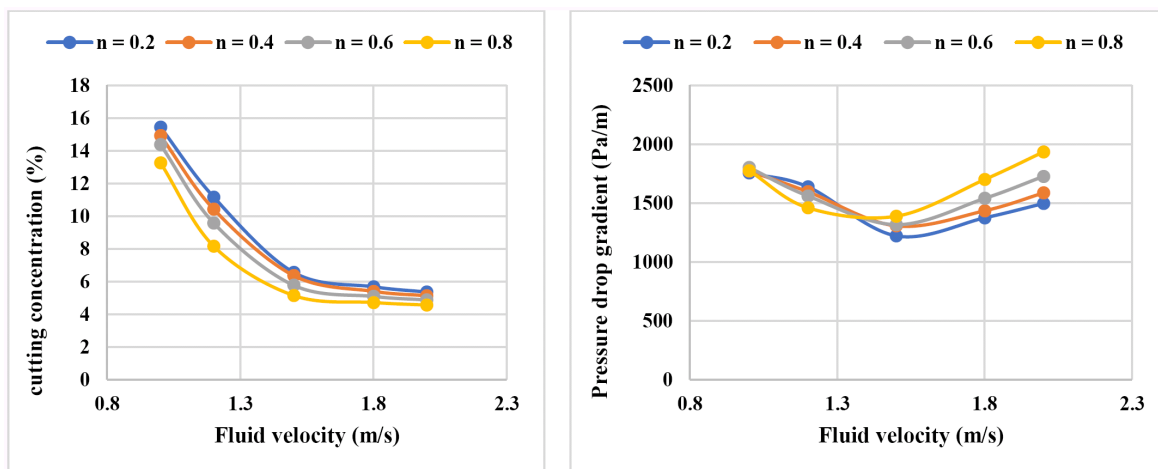


Figure 19—Influence of fluid velocity on cutting concentration and pressure gradient for various flow behavior indexes

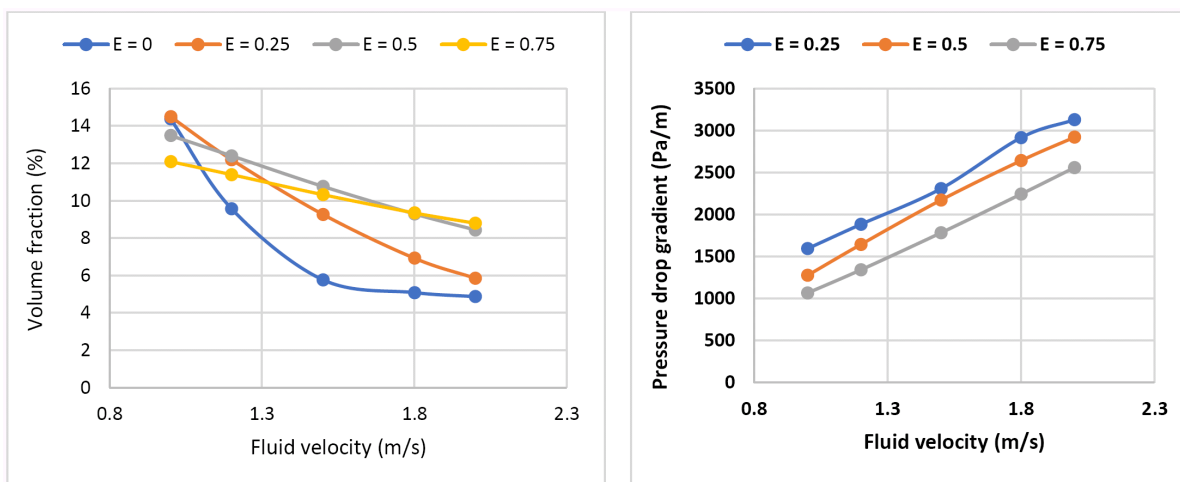


Figure 20—Influence of angular speed of the inner pipe on volume fraction and pressure gradient for various eccentricities

Real-Time Hole Cleaning Advisory

The Real-Time Hole Cleaning Advisory System provides an intuitive, color-coded risk visualization (Table 7, Figure 21) for critical drilling parameters, enabling immediate, informed decision-making. It categorizes risk for CTR_m, CCA_m, HCI, and TI_m (Red: High Risk; Yellow: Acceptable/Monitor; Green: Optimized).

Table 7—Real time hole cleaning advisory recommendations

Case	CCA _m		HCI		CTR _m		TI _m		Case Modified
1									Optimized
2									Increase GPM, Increase ROP
3									Stop Drilling, Condition hole & mud
4				Or		Or		Or	Control ROP
5		Or				Or		Or	Optimize rheological properties
6		Or		Or				Or	increase GPM
7		Or		Or					Optimize rheological properties
10						Or		Or	Control ROP, Optimize rheological properties
11				Or				Or	Control ROP, increase GPM
8				Or		Or			Control ROP, Optimize rheological properties
12		Or						Or	Optimize rheological properties, increase GPM,
9		Or				Or			Optimize rheological properties, increase GPM
13		Or		Or					Optimize rheological properties, increase GPM
14		Or							Optimize rheological properties, increase GPM, control ROP
15				Or					Control ROP, increase GPM, Optimize rheological properties
16						Or			Control ROP, increase GPM, Optimize rheological properties
17								Or	Control ROP, increase GPM, Optimize rheological properties

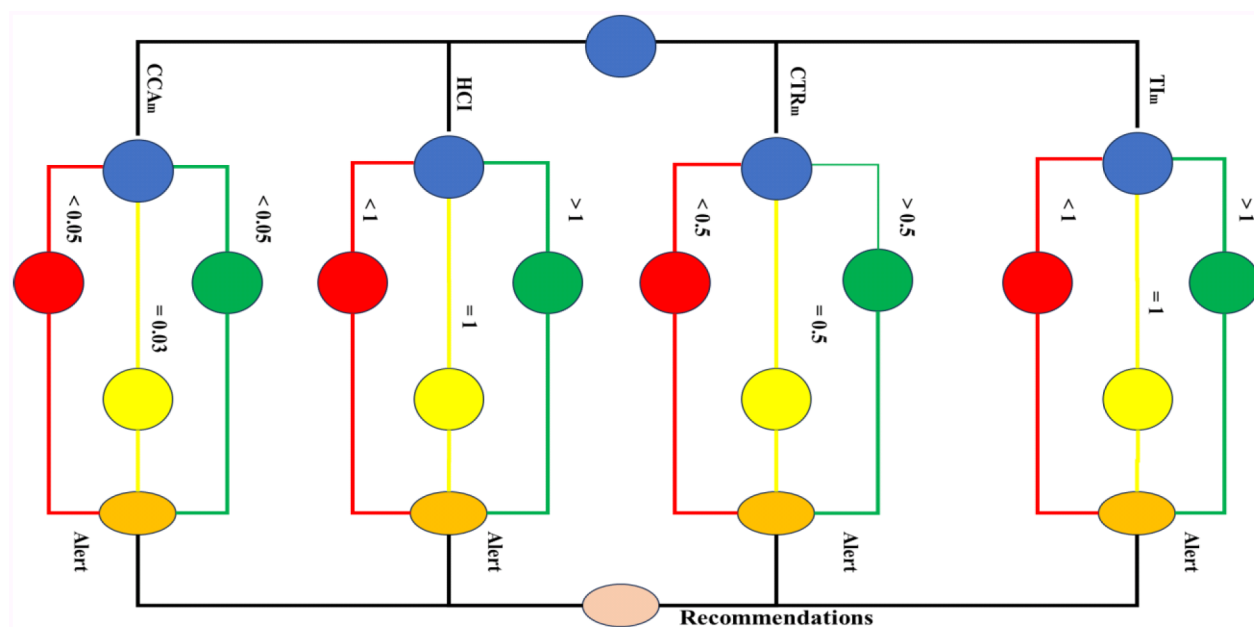


Figure 21—the real time hole cleaning advisory Alerts

System enhancements (Figure 22) include:

- Optimum GPM Values: Recommended flow rates for efficiency.
- Expanded Input Capability: Comprehensive data integration (liner/casing diameters, temperatures, connection time, RPMs) for enhanced accuracy.
- Dedicated Foreman Dashboard: Consolidated real-time view for rig-site decision-making.
- Intelligent GPM Adjustment Advice: Ensures efficient and safe operations.

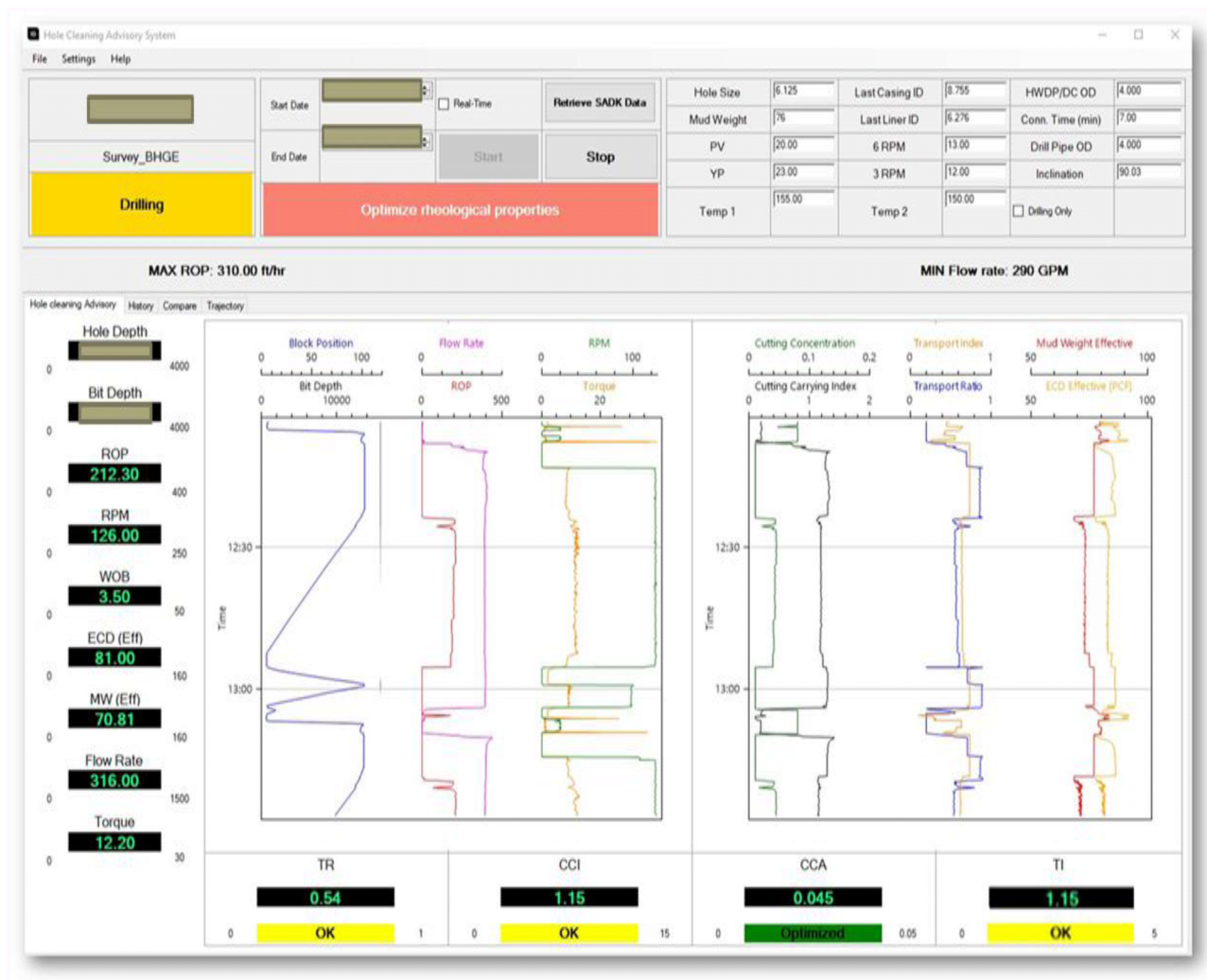


Figure 22—Real time Hole cleaning advisory system.

Field applications demonstrate remarkable results (Figure 22), facilitating record drilling rates (e.g., 5718 units at 77.79 fph), underscoring the system's exceptional efficiency in optimizing drilling processes for safer, more efficient, and cost-effective well delivery.

Conclusion and Recommendations

This work introduces a unique Hole Cleaning Advisory System, providing real-time performance indicators (TIm, HCI, CTRm, ECDeffc.m, MWeffc.m) validated against sensors (commercial, PWD) and AI. It significantly improves drilling performance.

Key Conclusions:

- **Comprehensive Automation:** TIm integrates dynamic mud weight, hydraulics, rheology, and cuttings for automated hole cleaning.
- **Quantifiable Efficiency:** TIm led to 56% ROP improvement and 44% torque reduction; HCI showed up to 50% drilling performance improvement; CTRm enhanced efficiency.
- **Superior Accuracy:** HCI outperformed CCI. ECDeffc.m/MWeffc.m showed exceptional accuracy (MAE 0.06-0.11%, MSE 0.08-0.29%), matching AI/commercial systems.
- **AI Validation:** ANN, SVM, DT accurately predicted ECD (99.63%) and MW (99.86%), reducing reliance on expensive equipment.

- Proactive Mitigation: The system identifies hole cleaning issues and stuck pipe risks in real-time, enhancing safety and efficiency.

Recommendations

1. Industry-Wide Adoption: Implement this validated system for standardized cleaning.
2. Continuous Enhancement: Integrate learning mechanisms for ongoing improvement.
3. Holistic Digitalization: Further integrate into broader drilling automation.
4. Training & Development: Ensure personnel leverage advanced capabilities for maximum benefit.

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